

MARHA MIDHATIEY

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Professional Summary

Experienced in Large Language Models (LLMs), neurosymbolic program synthesis, and visual reasoning. Also in model fine-tuning on HPC clusters, software engineering, database architecture, and leading academic and entrepreneurial field.

Technical Skills

Languages & Frameworks: Python, PyTorch, C++, C#, Java, PHP, Visual Basic, LaTeX, SQL

AI / Machine Learning: LLM Fine-Tuning (QLoRA, SFT, LoRA), Neurosymbolic AI, Computer Vision (CNNs, Transfer Learning), MelBandRoformer, Prompt Engineering

Databases & Tools: Oracle DB, MySQL, Power BI, Kafka, Gradio, RapidMiner, Git, Linux HPC Clusters

Technical Experience

Research Assistant (Data Science Lab) | GIST, South Korea (2024 – 2026)

- Researched hybrid AI architectures combining neurosymbolic methods and LLMs for visual reasoning using **Python** and **PyTorch**.
- Optimized fine-tuning of 7B/13B/32B LLMs on HPC clusters via **QLoRA, SFT, and LoRA**, significantly reducing memory footprint and accelerating training throughput.
- Authored research papers and formatted complex mathematical models and algorithms in **LaTeX** for journal and conference submissions.

System Analyst & Database Engineer (Intern) | 3F Resources Sdn. Bhd., Malaysia (2023)

- Developed government technology platforms for Ministry of Education using **Oracle DB, MySQL, Java, PHP, and Power BI**.
- Designed UI/UX and UML database structures, upgrading legacy systems to increase overall data processing efficiency.

System Administrator (Intern) | Rifhan Teknologi Sdn. Bhd., Malaysia (2020)

- Maintained high network availability across Ministry of Home Affairs, RMP HQ, and Ministry of Finance with **99% target uptime**. Managing IT desk ticket and enterprise monitoring tools.

Publications & Key Projects

- **Tracing and Correcting Programs (LLM Reasoning):** (*AAAI 2026 Bridge*) & *Master's Thesis*. Enhanced LLM multi-step reasoning using prompt signals and fine-tuning (**PyTorch, QLoRA**).
- **Driver Drowsiness Detection System:** *Best Paper Award (ISETech 2023)*. Developed real-time computer vision system via **Transfer Learning & CNNs**. Published in *Journal of Advanced Research in Applied Mechanics*.
- **Voice Synthesis & Sentiment Pipelines:** Built audio matching pipeline (**PyTorch, MelBandRoformer, Gradio, Kafka**) and conducted sentiment analysis on airline customer datasets using **RapidMiner**.

Education, Leadership & Community

Gwangju Institute of Science and Technology (GIST) — M.S. in Artificial Intelligence (Dept. of AI Convergence) | 2024 – Feb 2026

Universiti Teknologi MARA (UiTM) — B.S.(Hons.) in Intelligent Systems Engineering | GPA: 3.79/4.00 |

Diploma in Computer Science | GPA: 3.83/4.00

Honors & Awards: Vice Chancellor Award, Dean's Award (Every Semester), Best Paper Award (ISETech 2023).

Spoken Languages: English, Malay, Indonesian, Korean (Conversational).

- **Co-Lead, AI & Data Science Study Group (GIST):** Organized technical seminars, curated ML study materials, and managed venue logistics.
- **Founder & E-Commerce Operator:** Managed end-to-end global cross-border shipping logistics, digital marketing, and customer relationship management.
- **International Student Support Officer (GISA) & Volunteer Tutor:** Guided international student onboarding at GIST.
- **Volunteer Programming Tutor:** Mentored middle and high school students in introductory **Programming Language and Python** and algorithmic problem-solving.